P. Proof of Convergence for PASO with Sliding Window

P.1. Notation and Problem Restatement

Basic Notation. We denote the full sequence of weights up to time $T-1$ as $W = [\hat{w}_0, \ldots, \hat{w}_{T-1}]$. The norm $\|\cdot\|_F$ denotes the Frobenius norm. For model weights $w \in \mathbb{R}^n$ with $n > 1$, operations are assumed to be element-wise. The target fixed point $W^{seq} = [w_0^{seq}, w_1^{seq}, \ldots, w_{T-1}^{seq}]$ is the exact trajectory generated by standard autoregressive gradient descent.

The base update function for a single model state at index t is defined as:

$$F_{t-1}(\hat{w}_0, \cdots, \hat{w}_{t-1}) = \hat{w}_0 - \sum_{\tau=0}^{t-1} \eta_\tau g_\tau(\hat{w}_\tau, \ldots, \hat{w}_0) \tag{54}$$

We denote the prefix sequence up to time τ as $W_{\leq \tau} = [\hat{w}_0, \ldots, \hat{w}_\tau]$.

Block Decomposition & Core Variables. We systematically partition the total sequence into m consecutive functional blocks, $\mathcal{B}_1, \ldots, \mathcal{B}_m$, where $m \in [\lceil T/p \rceil, T]$.

- **Block Indices:** $\mathcal{B}_i = \{t_i, \ldots, t_{i+|\mathcal{B}_i|-1}\}$. The parallel capacity is strictly bounded by the maximum window size: $|\mathcal{B}_i| \leq p$.
- **Block State:** $W_{\mathcal{B}_i} = [\hat{w}_{t_i}, \ldots, \hat{w}_{t_{i+|\mathcal{B}_i|-1}}]$. The exact sequential target for this block is $W_{\mathcal{B}_i}^{seq}$.
- **Historical Context:** H_{i-1} is the actual historical input received by block i from previously processed blocks. The ideal sequential history is H_{i-1}^{seq}.
- **Block Operator:** For a fixed history H_{i-1}, the parallel evaluation mapping over the block at iteration j is defined as:

$$F_{\mathcal{B}_i}(W_{\mathcal{B}_i}^{(j-1)} \mid H_{i-1}) = \Big[F_{t_i}(\hat{w}_{t_i} \mid H_{i-1}), \ldots, F_{t_{i+|\mathcal{B}_i|-1}}(\hat{w}_{t_i}, \ldots, \hat{w}_{t_{i+|\mathcal{B}_i|-1}} \mid H_{i-1})\Big]$$

- **Block Error:** $E_i^{(j)} = \|W_{\mathcal{B}_i}^{(j)} - W_{\mathcal{B}_i}^{seq}\|_F$.

P.2. Assumptions.

Assumption P.1 (Lipschitz Continuity)**.** The gradient update function g_τ is uniformly Lipschitz continuous. There exists a constant $C > 0$ such that for any two sequences X_τ and Y_τ, $\|g_\tau(X_\tau) - g_\tau(Y_\tau)\|_F \leq C\|X_\tau - Y_\tau\|_F$.

Assumption P.2 (Bounded Initial Distance)**.** For any functional block i, the initial bounded drift before its localized parallel updates begin is bounded by a constant $D > 0$, defined as $\max_i \|W_{\mathcal{B}_i}^{(0)} - W_{\mathcal{B}_i}^{seq}\|_F \leq D$.

P.3. Main Theorem

Theorem P.3 (Convergence and Parallel Speedup of PASO)**.** *Consider a sequence of length T processed with a maximum sliding window size of $p \in [1, T]$, partitioned into m functional blocks. The system undergoes K total parallel iterations, yielding an average of K/m iterations per block. Suppose the update function satisfies Assumption P.1 and the initial drift is bounded by D (Assumption P.2).*

Let the internal block contraction factor ρ_1 and the historical propagation factor ρ_2 be conservatively bounded by the window size p across all blocks as:

$$\rho_1 = \sqrt{p} \cdot \max_i \sum_{\tau \in \mathcal{B}_i} \eta_\tau C \tag{55}$$

$$\rho_2 = \sqrt{p} \cdot \max_i \sum_{\tau \in H_{i-1}} \eta_\tau C \tag{56}$$

If the learning rate schedule ensures strict internal contraction such that $\rho_1 \in (0, 1)$, the PASO algorithm mathematically guarantees:

(1) Global Convergence Error Bound: *The final approximation error of the terminal block m is strictly bounded by:*

$$\|W_{\mathcal{B}_m}^{(K/m)} - W_{\mathcal{B}_m}^{seq}\|_F \leq D \cdot \rho_1^{\frac{K}{m}} \cdot \left(1 + \frac{\rho_2(1-\rho_1^{\frac{K}{m}})}{1-\rho_1}\right)^{m-1}$$

(2) Parallel Speedup Condition *($K < T$)**:** To achieve ϵ-precision convergence in strictly fewer iterations than standard sequential execution, the system must satisfy the condition at $K = T$:*

$$\rho_1^{\frac{T}{m}} \cdot \left(1 + \frac{\rho_2(1-\rho_1^{\frac{T}{m}})}{1-\rho_1}\right)^{m-1} < \frac{\epsilon}{D}$$

This mathematical threshold guarantees that the exponential internal decay ($\rho_1^{T/m}$) governed by the window capacity is sufficient to overpower the historical structural error aggregation.

Corollary P.4. *The decoupled bound established in Theorem P.3 serves as a strict generalization of the uniform global contraction mapping. If we consider the stricter condition where the global trajectory operates under a single uniform contraction factor $\rho \in (0, 1)$, we can symmetrically bound both the internal block mapping and the historical propagation by ρ (i.e., $\rho_1 \leq \rho$ and $\rho_2 \leq \rho$). Under this uniform global assumption, the historical multiplier term elegantly collapses via standard algebraic simplification:*

$$1 + \frac{\rho(1-\rho^{\frac{K}{m}})}{1-\rho} = \frac{1-\rho+\rho-\rho^{\frac{K}{m}+1}}{1-\rho} = \frac{1-\rho^{\frac{K}{m}+1}}{1-\rho} \tag{57}$$

Substituting this collapsed term back into the global convergence error bound strictly reduces our generalized result to the simplified uniform bound:

$$\|W_{\mathcal{B}_m}^{(K/m)} - W_{\mathcal{B}_m}^{seq}\|_F \leq D \cdot \rho^{\frac{K}{m}} \cdot \left(\frac{1-\rho^{\frac{K}{m}+1}}{1-\rho}\right)^{m-1} \tag{58}$$

P.4. Phase 1: Finite Bounds on Iteration Count

Lemma P.5 (Finite Bounds of Termination)**.** *For the PASO algorithm to iteratively process the trajectory via sliding windows, the structural hyperparameters and resulting iteration counts are bounded by $\lceil T/p \rceil \leq K \leq T$ and $\lceil T/p \rceil \leq m \leq T$.*

Proof. The causal triangularity inherently forces the first element of any active window to reach its fixed point in exactly one iteration, yielding a successive difference of 0. Because adaptive tolerance $\delta^{(k)} \geq 0$, the condition $0 \leq \delta^{(k)}$ structurally guarantees a minimum forward stride of $s_k \geq 1$. Thus, the algorithm naturally terminates in at most T iterations ($K \leq T$). Conversely, since the maximum advance step is upper-bounded by the window capacity ($s_k \leq p$), completing the sequence requires $\sum_{k=1}^{K} s_k = T \implies K \cdot p \geq T$, establishing $K \geq \lceil T/p \rceil$. The number of structural blocks m is analogously bounded. □

P.5. Phase 2: Block-wise Contraction Mapping

Lemma P.6 (Block-wise Contraction Mapping)**.** *For any functional block $\mathcal{B}_i$ and fixed historical input H_{i-1}, the operator $F_{\mathcal{B}_i}(\cdot \mid H_{i-1})$ forms a strict contraction mapping bounded by ρ_1.*

Proof. Let W and V be two distinct sequence states for block $\mathcal{B}_i$. Since history H_{i-1} is identical, gradient updates prior to index t_i perfectly cancel out. For any specific index $t \in \mathcal{B}_i$:

$$F_t(W \mid H_{i-1}) - F_t(V \mid H_{i-1}) = -\sum_{\tau=t_i}^{t} \eta_\tau \left(g_\tau(W) - g_\tau(V)\right) \tag{59}$$

By Assumption P.1, because W and V share the identical history prefix, the gradient difference is strictly bounded by their distance across the current block:

$$\|g_\tau(W) - g_\tau(V)\|_F \leq C\|W_{\leq\tau} - V_{\leq\tau}\|_F \leq C\|W_{\mathcal{B}_i} - V_{\mathcal{B}_i}\|_F \tag{60}$$

Applying the triangle inequality to the state update and substituting this bound:

$$\|F_t(W \mid H_{i-1}) - F_t(V \mid H_{i-1})\|_F \leq \left(\sum_{\tau=t_i}^{t} \eta_\tau C\right) \|W_{\mathcal{B}_i} - V_{\mathcal{B}_i}\|_F \tag{61}$$

Since the learning rate and Lipschitz constant are non-negative, we can strictly bound the partial sum up to index t by extending it to the entire block $\mathcal{B}_i$. By definition of the Frobenius norm over the evaluated block (bounded by capacity p):

$$\begin{aligned}\|F_{\mathcal{B}_i}(W \mid H_{i-1}) - F_{\mathcal{B}_i}(V \mid H_{i-1})\|_F &= \sqrt{\sum_{t\in\mathcal{B}_i} \|F_t(W\dots) - F_t(V\dots)\|_F^2} \\ &\leq \sqrt{\sum_{k=0}^{p-1} \left(\sum_{\tau\in\mathcal{B}_i} \eta_\tau C\right)^2} \cdot \|W_{\mathcal{B}_i} - V_{\mathcal{B}_i}\|_F \\ &= \left(\sqrt{p} \sum_{\tau\in\mathcal{B}_i} \eta_\tau C\right) \|W_{\mathcal{B}_i} - V_{\mathcal{B}_i}\|_F \end{aligned} \tag{62}$$

By the definition in Theorem P.3, this bounding coefficient is exactly ρ_1. Because the learning rate schedule ensures $\rho_1 < 1$, it mathematical guarantees a strict internal contraction mapping. □

P.6. Phase 3: Global Convergence Error Bound

Proof of Theorem P.3. **Step 1: Single Iteration Recurrence & Historical Expansion.** Consider block i at iteration $j \geq 1$. Bounding the target distance via the triangle inequality with a cross-term:

$$E_i^{(j)} \leq \underbrace{\|F_{\mathcal{B}_i}(W_{\mathcal{B}_i}^{(j-1)} \mid H_{i-1}) - F_{\mathcal{B}_i}(W_{\mathcal{B}_i}^{seq} \mid H_{i-1})\|_F}_{\text{Internal Contraction: } \leq \rho_1 E_i^{(j-1)}} + \underbrace{\|F_{\mathcal{B}_i}(W_{\mathcal{B}_i}^{seq} \mid H_{i-1}) - F_{\mathcal{B}_i}(W_{\mathcal{B}_i}^{seq} \mid H_{i-1}^{seq})\|_F}_{\text{Historical Error Propagation}} \tag{63}$$

For the historical term, internal gradient contributions ($\tau \geq t_i$) cancel out since the target block states $W_{\mathcal{B}_i}^{seq}$ are identical. The residual difference is driven purely by the history $\tau \in H_{i-1}$:

$$\|F_t(\cdots \mid H_{i-1}) - F_t(\cdots \mid H_{i-1}^{seq})\|_F \leq \left(\sum_{\tau\in H_{i-1}} \eta_\tau C\right) \|H_{i-1} - H_{i-1}^{seq}\|_F \tag{64}$$

Aggregating via the root sum of squares over the parallel components, and explicitly bounding the dynamic block size by the window capacity ($|\mathcal{B}_i| \leq p$), we obtain the p-dependent historical bound ρ_2:

$$\begin{aligned}\|F_{\mathcal{B}_i}(\dots) - F_{\mathcal{B}_i}^{seq}(\dots)\|_F &\leq \left(\sqrt{|\mathcal{B}_i|} \sum_{\tau\in H_{i-1}} \eta_\tau C\right) \|H_{i-1} - H_{i-1}^{seq}\|_F \\ &\leq \left(\sqrt{p} \sum_{\tau\in H_{i-1}} \eta_\tau C\right) \|H_{i-1} - H_{i-1}^{seq}\|_F \\ &\leq \rho_2 \|H_{i-1} - H_{i-1}^{seq}\|_F \end{aligned} \tag{65}$$

Letting $\delta_{hist} = \rho_2 \|H_{i-1} - H_{i-1}^{seq}\|_F$, the single iteration recurrence strictly resolves to:

$$E_i^{(j)} \leq \rho_1 E_i^{(j-1)} + \delta_{hist} \tag{66}$$

Step 2: Unrolling Iterations within the Block. For block i processing over K/m iterations, unrolling the linear recurrence yields:

$$E_i \leq \rho_1^{\frac{K}{m}} E_i^{(0)} + \frac{1 - \rho_1^{\frac{K}{m}}}{1 - \rho_1} \delta_{hist} \tag{67}$$

The initial error is bounded by D. The historical delta concatenates previous block errors $\delta_{hist} \leq \rho_2 \sum_{k=1}^{i-1} E_k$, establishing the strictly decoupled causal propagation:

$$E_i \leq \rho_1^{\frac{K}{m}} D + \frac{\rho_2(1-\rho_1^{\frac{K}{m}})}{1-\rho_1} \sum_{k=1}^{i-1} E_k \tag{68}$$

Step 3: Solving the Global Propagation to the Terminal Block. Let constants $A = \rho_1^{\frac{K}{m}} D$ and $B = \frac{\rho_2(1-\rho_1^{\frac{K}{m}})}{1-\rho_1}$. The propagation strictly follows the recursive sequence:

$$E_i \leq A + B \sum_{k=1}^{i-1} E_k \tag{69}$$

We solve this recurrence strictly using strong mathematical induction to prove that the closed-form solution is $E_i \leq A(1+B)^{i-1}$ for all $i \geq 1$.

Base Case ($i = 1$): The historical summation is empty, yielding $E_1 \leq A = A(1+B)^0$, which holds true trivially.

Inductive Step: Assume the bound $E_k \leq A(1+B)^{k-1}$ holds for all previous blocks $k \in [1, i-1]$. We evaluate the error for block i:

$$\begin{aligned} E_i &\leq A + B \sum_{k=1}^{i-1} E_k \\ &\leq A + B \sum_{k=1}^{i-1} A(1+B)^{k-1} \end{aligned} \tag{70}$$

Recognizing that the summation forms a standard geometric series with a common ratio of $(1+B)$, we can evaluate it exactly:

$$\sum_{k=1}^{i-1} (1+B)^{k-1} = \frac{(1+B)^{i-1} - 1}{(1+B) - 1} = \frac{(1+B)^{i-1} - 1}{B} \tag{71}$$

Substituting this geometric sum back into the inequality effortlessly cancels out the B multiplier:

$$\begin{aligned} E_i &\leq A + B \cdot A \left(\frac{(1+B)^{i-1} - 1}{B} \right) \\ &= A + A\left((1+B)^{i-1} - 1\right) \\ &= A(1+B)^{i-1} \end{aligned} \tag{72}$$

This flawlessly concludes the induction. Evaluating this closed-form solution for the terminal functional block ($i = m$) provides $E_m \leq A(1+B)^{m-1}$.

Finally, substituting the original system variables A and B back into this expression yields the exact global convergence bound parameterized by the window size p:

$$\|W_{\mathcal{B}_m}^{(K/m)} - W_{\mathcal{B}_m}^{seq}\|_F \leq D \cdot \rho_1^{\frac{K}{m}} \cdot \left(1 + \frac{\rho_2(1-\rho_1^{\frac{K}{m}})}{1-\rho_1} \right)^{m-1} \tag{73}$$

Step 4: Conditions for Parallel Speedup ($K < T$). Because the internal mapping is a strict contraction ($\rho_1 \in (0,1)$), the exponential decay of the leading factor $\rho_1^{\frac{K}{m}}$ strictly dominates the history aggregation as $K \to \infty$. The global error bound monotonically decreases with respect to iterations K.

Achieving ϵ-precision in strictly fewer iterations than T is mathematically guaranteed if and only if the error bound evaluated at exactly $K = T$ is less than ϵ. Substituting $K = T$ yields:

$$D \cdot \rho_1^{\frac{T}{m}} \cdot \left(1 + \frac{\rho_2(1-\rho_1^{\frac{T}{m}})}{1-\rho_1} \right)^{m-1} < \epsilon \tag{74}$$

Dividing by D isolates the threshold requirement for early convergence:

$$\rho_1^{\frac{T}{m}} \cdot \left(1 + \frac{\rho_2(1-\rho_1^{\frac{T}{m}})}{1-\rho_1}\right)^{m-1} < \frac{\epsilon}{D} \tag{75}$$

□